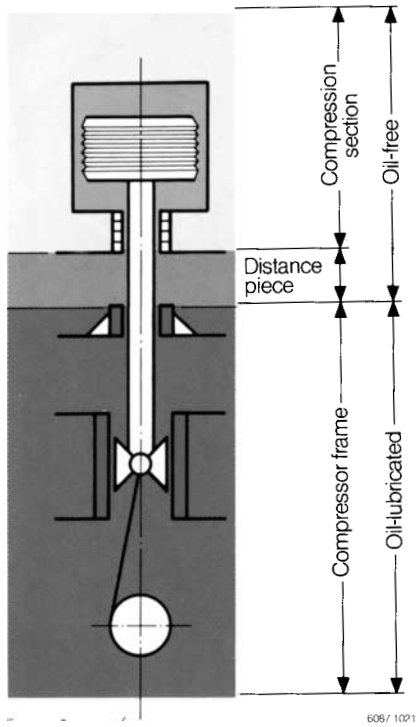



**FIG. C-101 Typical section showing oil-lubricated and oil-free zones. (Source: Sulzer-Burckhardt.)**

for applications where the process gas is heavily contaminated with impurities, such as polymerization products or other very small and hard particles. They have effectively no influence on the labyrinth seal performance, compressor reliability, wear rate, and maintenance intervals.

Piston and piston rod are guided by the crosshead and the guide bearing, which are located in the oil-lubricated crankcase. Both guiding elements are made of metal and are oil lubricated, thus ensuring a precisely linear operation of the labyrinth piston as well as an extremely long life of the piston/piston rod guiding system.

The distance piece separates the gas compressing section from the oil-lubricated crankcase.

## LABY® design options and features

A large variety of standard labyrinth-piston compressors, with many additional cylinder blocks, is available with suction volumes up to 11,000 $m^3/h$ and discharge pressures exceeding 300 bar. See Figs. C-102 through C-104 for various types. Design features of the totally closed "K"-type compressor with gas- and pressure-tight crankcase are illustrated in Fig. C-103C and D. See these figures and Fig. C-105.

### Common features

#### The labyrinth piston (see Fig. C-106)

- May be double- or single-acting (depending on application)
- Seals by repeated gas throttling

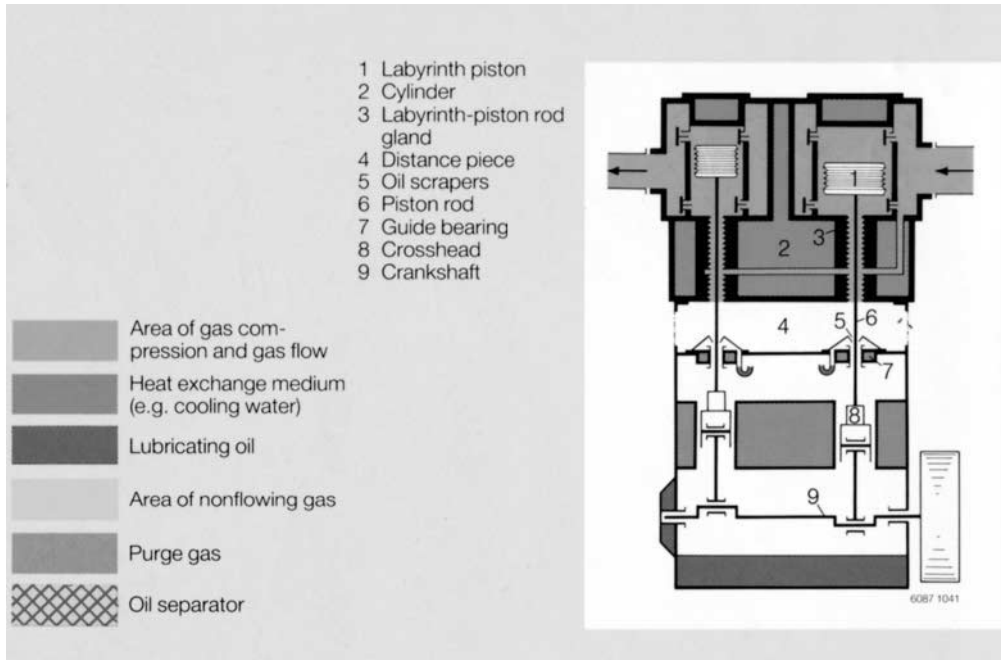



**FIG. C-102A With open distance piece: This standard compressor is equipped with an open distance piece and a nonpressurized crankcase. It is used for compression of gases, where a strict separation between cylinder and crankcase is essential and where process gas is permitted in the open distance piece (e.g., for $O_2$, $N_2$, $CO_2$, process air, etc., generally in the industrial gas industry). (Source: Sulzer-Burckhardt.)**

- Consists of a very small number of parts
- Is made of solid metal without any plastic material
- Avoids permanent mechanical friction
- Avoids contamination or fouling of the process gas
- Guarantees extremely long sealing element life and assures low maintenance cost
- Accepts a wide range of operating temperatures (–160 to +270°C and higher)
- Is insensitive to impurities in the gas
- Ensures unexceeded reliability in oxygen service

**The compressor valve (see Fig. C-107)**

- Helps to achieve ideal combinations of cylinder design, valve size, and compressor plant components
- Ensures high reliability and availability of the compressor
- Embodies frictionless guided plates with very low lift and extremely good fatigue properties
- Consists of identical parts for suction and discharge side, but special design features prevent inadvertent wrong assembly of valves into the cylinder
- Comprises stationary parts, such as valve seat and stroke limiter not being cast, but machined out of special stainless steel

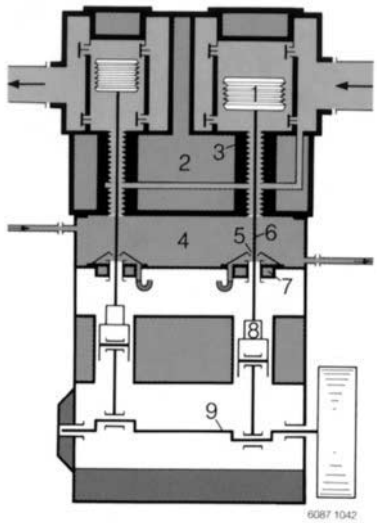



**FIG. C-102B With closed and purged distance piece: The distance piece of the standard open-type compressor is closed and purged with nitrogen, air, or another suitable gas. It is used for compression of gases, where a strict separation between cylinder and crankcase is essential and where no process gas may leak to the surroundings or no ambient air may enter the distance piece (e.g., for weather protection). (Source: Sulzer-Burckhardt.)**

- Incorporates dynamically moving parts, such as valve plates and damper plates, manufactured according to most modern techniques and made of special stainless steel
- Includes aerodynamically optimal shapes with low pressure drop

**The piston rod gland (see Fig. C-108)**

- Features radially floating and self-centering labyrinth sealing rings made of graphite
- Comprises stainless-steel gland chambers for the sealing rings
- Incorporates a leak-gas collecting chamber at the lower end for feeding the leak gas, where possible, back to the suction-side first stage
- Allows, if necessary, for specially designed applications with several connections

**The piston rod guide bearing (see Fig. C-109)**

- Is available in cooled or uncooled application
- Is available with a replaceable bush and is splash-lubricated
- Is combined with the oil scrapers and designed to exclude oil from the distance piece, the piston rod gland area, or the cylinder section. It eliminates the necessity for additional oil-vapor removal equipment.

**The crankshaft seal for open-type compressors (see Fig. C-110)**

- Is equipped with an oil slinger ring and, except on the smaller compressors, with an additional packing ring
- Is designed to exclude dust and dirt from the crankcase and to provide an oil-tight crankshaft passage through the crankcase wall

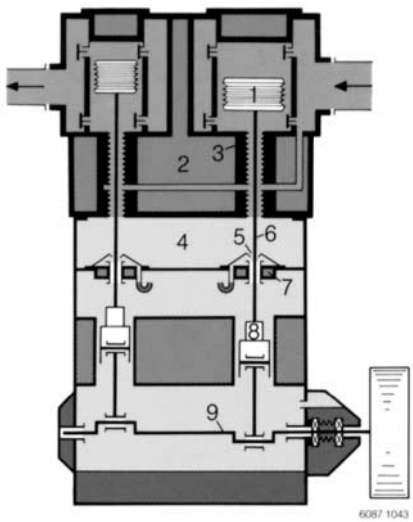



**FIG. C-102C With gas-tight crankcase and mechanical crankshaft seal: The distance piece of the standard open-type compressor is closed, and the crankshaft bears a mechanical gas-tight seal where it passes through the crankcase wall. This design is used for compression of gases which are compatible with the lubricating oil (e.g., for hydrocarbon gases, CO, He, $H_2$, Ar, etc.) and where no process gas may leak to the surroundings. The suction pressure is limited by the design pressure of the crankcase. (Source: Sulzer-Burckhardt.)**

**The crankshaft seal for compressors with gas-tight crankcase (see Fig. C-111)**

- Is equipped with a mechanical sealing system completely immersed in lubricating oil
- Prevents oil from draining into the crankcase during standstill periods
- Incorporates an additional sealing ring to exclude dust and dirt from the shaft seal area
- Provides and ensures a gas- and oil-tight crankshaft passage through the crankcase wall

### LABY® research and development projects

**The compressor valves.** Valves installed in a reciprocating compressor have a tough life. They have to open and close automatically once every crankshaft revolution, quickly and reliably under severe temperature and pressure conditions. Troubles and excessive wear or losses are avoidable if valve quality as well as the match of valve, compressor, and operating conditions are optimal. See Fig. C-112.

Painstaking design and years of feedback from operations have raised compressor valves to a very high standard in terms of material, manufacturing technology, and aerodynamic shape. Nevertheless, to retain an OEM's lead in valve technology, continue to invest in valve research. There are further possibilities to reduce stress peaks in dynamically loaded parts, to optimize the aerodynamic characteristics, to influence the movement of the dynamic parts, to introduce improved materials and manufacturing techniques.

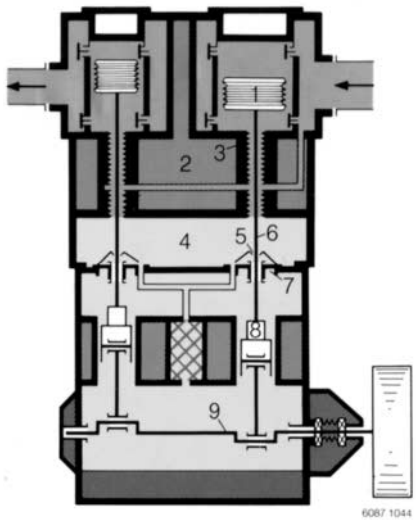



**FIG. C-102D With gas- and pressure-tight crankcase and mechanical crankshaft seal: This standard compressor is equipped with a closed single-piece crankcase designed for a gas pressure of 15 bar or higher. All openings are closed and sealed with o-rings. The crankshaft bears a mechanical gas-tight seal where it passes through the crankcase wall. Since the crankcase is filled with process gas, this machine is used to compress gases that are compatible with the lubricating oil and where no process gas may leak to the surroundings. Suction pressure may range between subatmospheric and crankcase design pressure. This machine finds its applications in closed cycles, for hydrocarbon gases, refrigerants, VCM, CO, $N_2$, $CO_2$, He, $H_2$, Ar, etc. (Source: Sulzer-Burckhardt.)**

**The sealing labyrinth.** This component is the subject of ongoing research. Considerable time and effort is invested in exploring the flow behavior of gases in oscillating sealing labyrinths and comparing the results with simulated computer calculations. Better understanding of the influence of piston speed, labyrinth shape, labyrinth clearance, and other factors on compressor performance emerges. See Figs. C-111 through C-113.

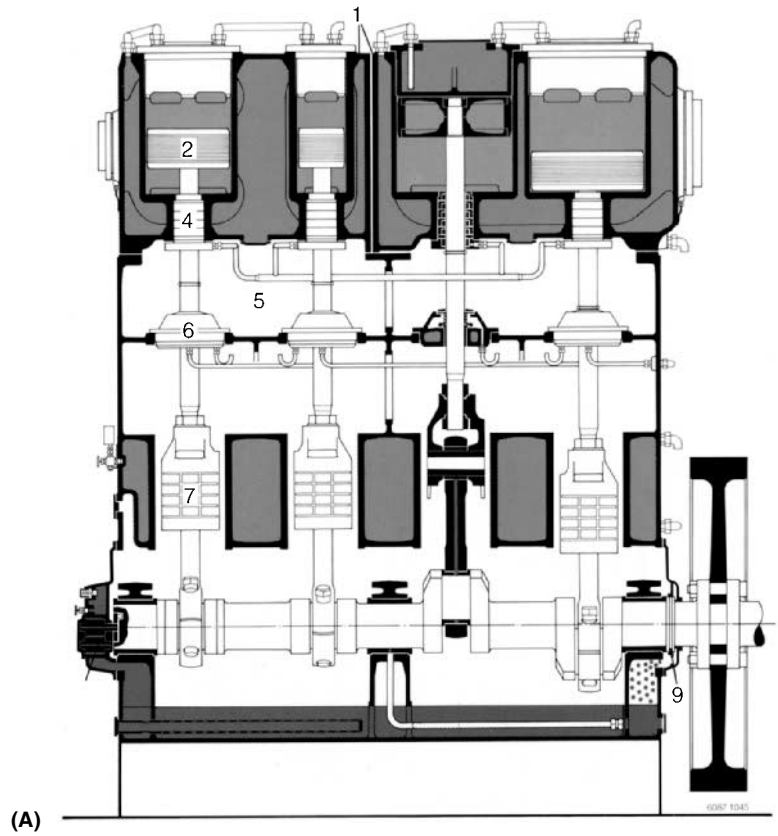
**Dynamic crank throw behavior.** This remains a subject of investigation. Compressor parts are not entirely rigid, but rather flexible, and may oscillate or vibrate during operation. It is important to have fundamental and detailed knowledge of means to eliminate or suppress undesired movements. See Figs. C-114 and C-115.

**Acoustic calculations.** Nowadays these are important. With computer technology one can choose between both digital and analog studies according to API 618.

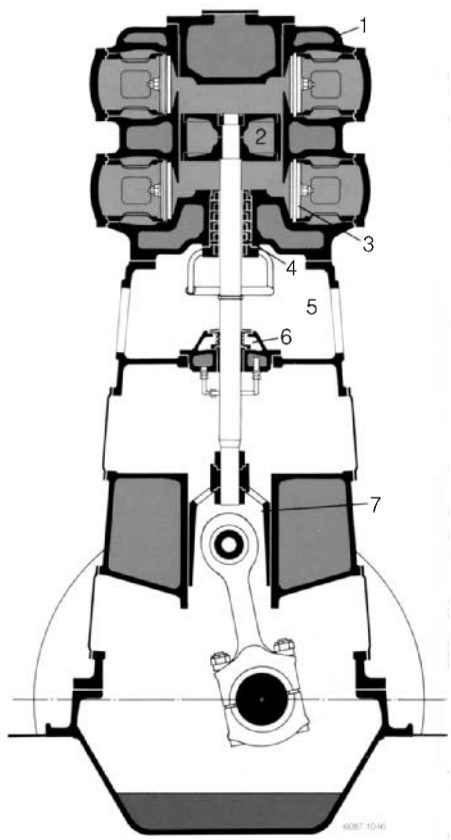
**New design materials.** For cylinders, pistons, piston rods, and other parts, new design materials are under consideration to meet new requirements from customers or to employ labyrinth-piston compressor in new applications.

**Quality control (QC) for reciprocating compressors.** QC is regularly adapted to possible new requirements of the market as well as to new measuring and monitoring methods to obtain optimal quality.

Quality inspection is performed during the manufacturing process after each important step. All pressure-stressed parts, such as cylinders, cylinder covers,

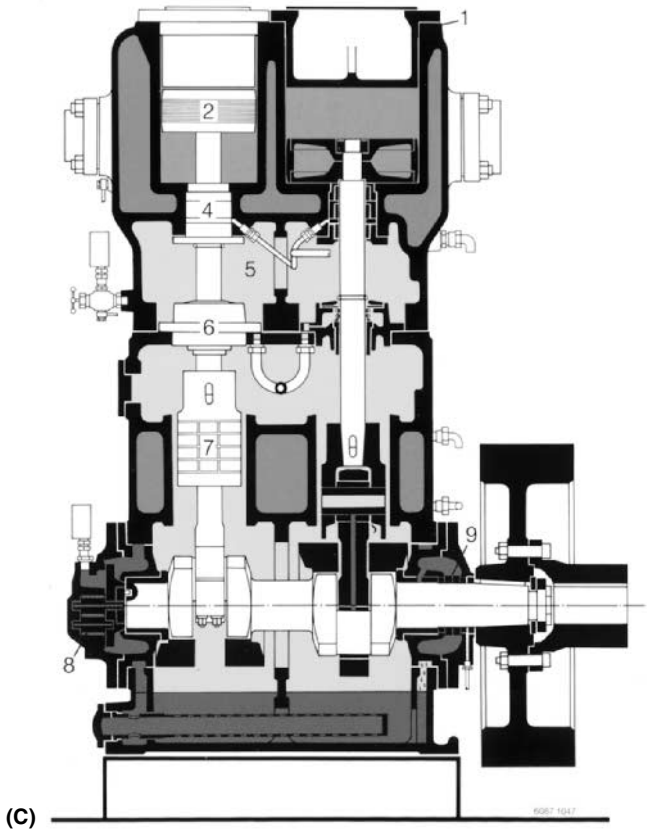



(A)

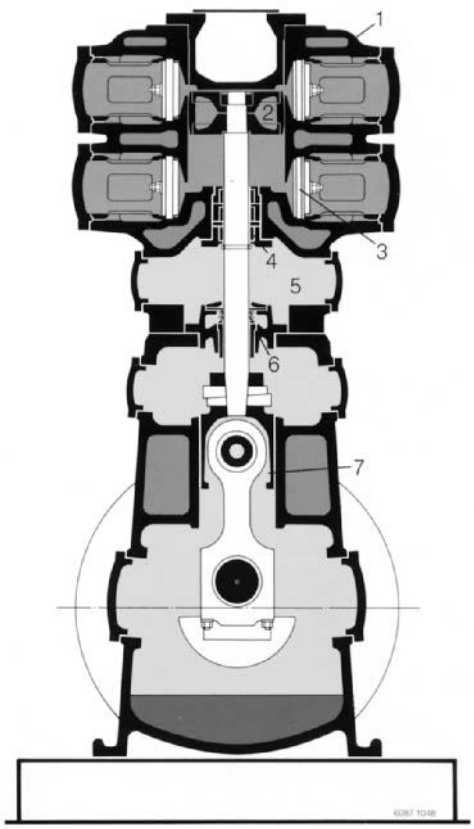



(B)

**FIG. C-103A, B Design features of D- and E-type compressors with open distance piece. (Source: Sulzer-Burckhardt.)**




(C)

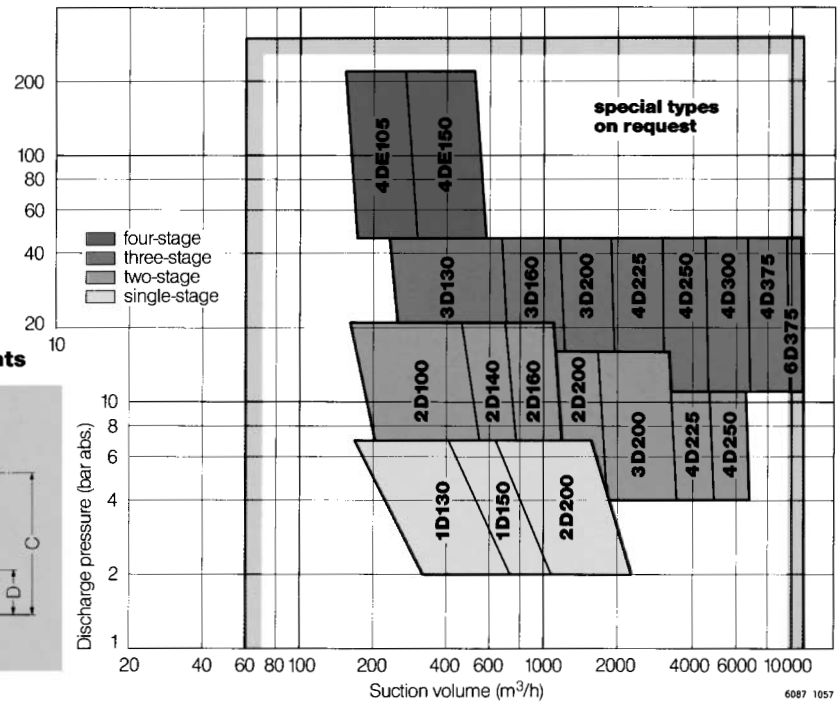



(D)

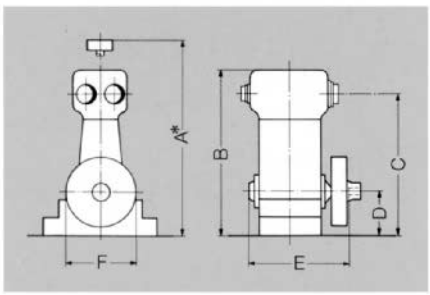
**FIG. C-103C, D Design features of the totally closed K-type compressor with gas- and pressure-tight crankcase. (Source: Sulzer-Burckhardt.)**

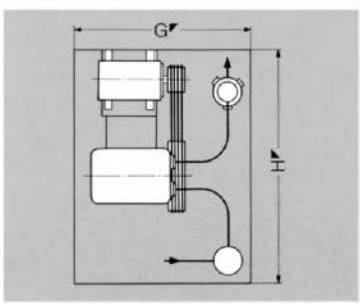
Performance range based on diatomic gases with molecular mass between 15 and 50 kg/kmol

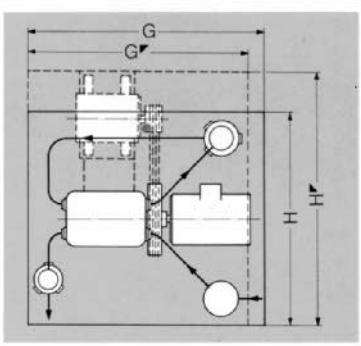
Volume flow based on suction condition 1 bar abs, 20 °C

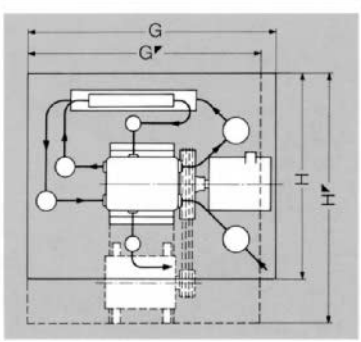



## Principal dimensions and weights




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**Single-stage compressors**

| Type | Dimensions A* mm | B mm | C mm | D mm | E mm | F mm | Compr. weight kg | Foundation** G* mm | H* mm |
|---|---|---|---|---|---|---|---|---|---|
| 1D130-1A | 3465 | 2385 | 2010 | 500 | 765 | Ø 910 | 1900 | 2500 | 3300 |
| 1D150-1A | 3710 | 2570 | 2185 | 550 | 870 | Ø1000 | 2500 | 2800 | 3600 |
| 2D200-1A | 4450 | 3090 | 2597 | 700 | 1510 | Ø1300 | 5200 | 3200 | 4200 |

**Two-stage compressors**

| Type | Dimensions A* mm | B mm | C mm | D mm | E mm | F mm | Compr. weight kg | Foundation** G mm | H mm | Foundation** G* mm | H* mm |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2D100-2A | 2740 | 1975 | 1585 | 430 | 970 | Ø 770 | 1800 | 3500 | 2500 | 2800 | 3200 |
| 2D140-2A | 3160 | 2305 | 1845 | 500 | 1110 | Ø 900 | 2500 | 4000 | 2800 | 3200 | 3800 |
| 2D160-2A | 3510 | 2550 | 2060 | 550 | 1220 | Ø1000 | 3200 | 4500 | 3200 | 3500 | 4400 |
| 2D200-2A | 4400 | 3090 | 2600 | 700 | 1510 | Ø1200 | 5600 | 4500 | 3500 | – | – |
| 3D200-2A | 4230 | 3025 | 2530 | 700 | 2200 | Ø1200 | 7200 | 6000 | 4000 | – | – |
| 4D225-2A | 4500 | 3210 | 2695 | 700 | 2605 | Ø1200 | 11700 | 7500 | 5000 | – | – |
| 4D250-2B | 5000 | 3670 | 3100 | 800 | 3005 | Ø1500 | 15800 | 8500 | 5500 | – | – |

**Three-stage compressors**

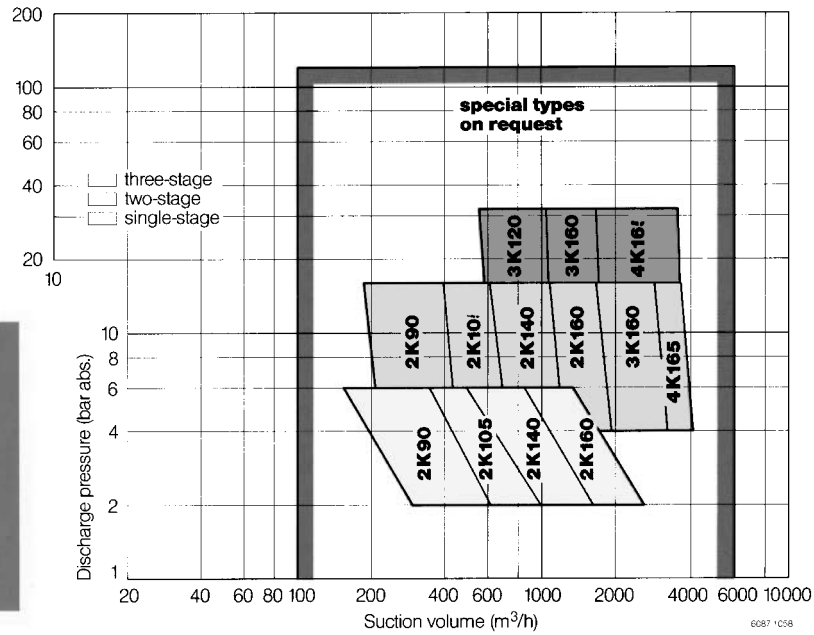
| Type | Dimensions A* mm | B mm | C mm | D mm | E mm | F mm | Compr. weight kg | Foundation** G mm | H mm | Foundation** G* mm | H* mm |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 3D130-3A | 3300 | 2325 | 1910 | 500 | 1345 | Ø 900 | 3500 | 4500 | 3200 | 4000 | 3800 |
| 3D160-3A | 3560 | 2825 | 2110 | 600 | 1470 | Ø1000 | 4500 | 5000 | 4000 | 4300 | 4800 |
| 3D200-3A | 4230 | 3010 | 2530 | 700 | 2200 | Ø1200 | 7200 | 7000 | 4500 | – | – |
| 4D225-3A | 4500 | 3210 | 2695 | 700 | 2605 | Ø1300 | 11600 | 8000 | 5000 | – | – |
| 4D250-3A | 5000 | 3670 | 3105 | 800 | 3005 | Ø1500 | 16400 | 9000 | 6000 | – | – |
| 4D300-3A | 6000 | 4315 | 3595 | 1000 | 3515 | Ø1800 | 24600 | 10000 | 7000 | – | – |
| 4D375-3A | 6750 | 4950 | 4075 | 1050 | 4210 | Ø2000 | 39800 | 11000 | 8000 | – | – |

* Required height to remove piston and piston rod  ** Foundation dimensions are for estimates only

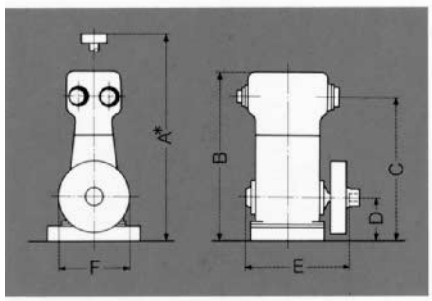
**FIG. C-104 Dimensions and performance parameters for D- and E-type compressors. (Source: Sulzer-Burckhardt.)**

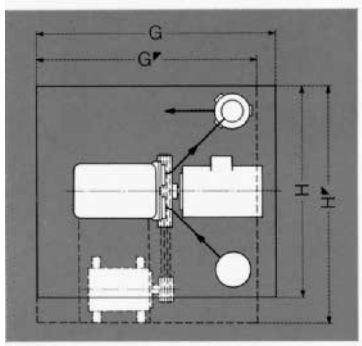
Performance range based on diatomic gases with molecular mass between 15 and 50 kg/kmol

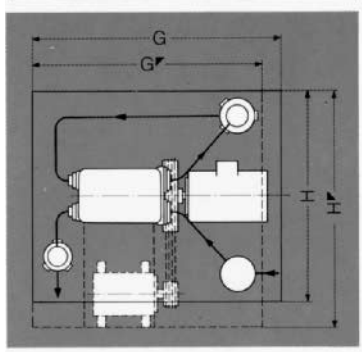
Volume flow based on suction condition 1 bar abs, 20°C

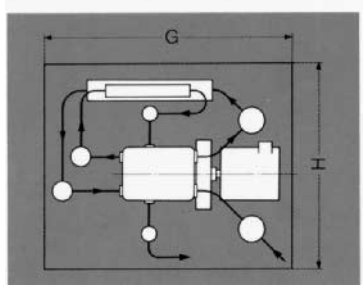



## Principal dimensions and weights




**Single-stage compressors**

| Type | Dimensions A* mm | B mm | C mm | D mm | E mm | F mm | Compr. weight kg | Foundation** G mm | H mm | Foundation** G* mm | H* mm |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2K 90-1A | 2260 | 1655 | 1420 | 520 | 1010 | Ø 600 | 1600 | 3200 | 2500 | 2800 | 3200 |
| 2K105-1A | 2640 | 1895 | 1615 | 570 | 1165 | Ø 660 | 2300 | 3800 | 2800 | 3200 | 3800 |
| 2K140-1A | 3180 | 2325 | 1985 | 650 | 1365 | Ø 900 | 4000 | 4200 | 3200 | 3500 | 4400 |
| 2K160-1A | 3470 | 2635 | 2155 | 710 | 1520 | Ø1000 | 5500 | 5000 | 3500 | – | – |

**Two-stage compressors**

| Type | Dimensions A* mm | B mm | C mm | D mm | E mm | F mm | Compr. weight kg | Foundation** G mm | H mm | Foundation** G* mm | H* mm |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2K 90-2A | 2270 | 1710 | 1420 | 520 | 1000 | Ø 600 | 1700 | 3200 | 2800 | 3000 | 3500 |
| 2K105-2A | 2640 | 1915 | 1615 | 570 | 1165 | Ø 660 | 2500 | 4200 | 3000 | 3800 | 4000 |
| 2K120-2A | 3150 | 2360 | 1870 | 630 | 1250 | Ø 900 | 2600 | 4500 | 3500 | 3800 | 4500 |
| 2K140-2A | 3200 | 2360 | 1985 | 650 | 1365 | Ø 900 | 4300 | 4500 | 3500 | 4000 | 4500 |
| 2K160-2A | 3470 | 2635 | 2155 | 710 | 1520 | Ø1000 | 5900 | 5000 | 4000 | – | – |
| 3K160-2C | 3580 | 2685 | 2210 | 710 | 2175 | Ø1000 | 8200 | 6000 | 4200 | – | – |
| 4K165-2C | 3760 | 2800 | 2320 | 760 | 2975 | Ø1000 | 14100 | 7500 | 4500 | – | – |

**Three-stage compressors**

| Type | Dimensions A* mm | B mm | C mm | D mm | E mm | F mm | Compr. weight kg | Foundation** G mm | H mm |
|---|---|---|---|---|---|---|---|---|---|
| 3K120-3E | 3100 | 2290 | 1900 | 630 | 1750 | Ø 900 | 5700 | 5000 | 4000 |
| 3K160-3E | 3150 | 2630 | 2170 | 670 | 2175 | Ø1000 | 8200 | 7000 | 4800 |
| 4K165-3E | 3760 | 2800 | 2320 | 760 | 2975 | Ø1000 | 14100 | 8000 | 5000 |

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* Required height to remove piston and piston rod ** Foundation dimensions are for estimates only

**FIG. C-105 Dimensions and performance parameters for K-type compressor. (Source: Sulzer-Burckhardt.)**

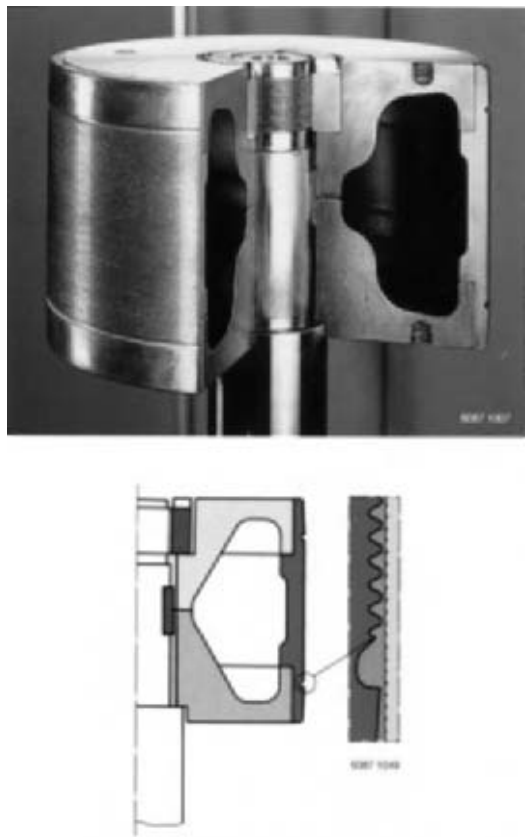


FIG. C-106 The labyrinth piston. (Source: Sulzer-Burckhardt.)

crankcases, oil-pump casings, and others, are hydraulically tested, and leakage tests are made on the assembled gas- and pressure-tight compressors. During assembly, all bearing and piston clearances are measured and recorded on request, and the alignment is checked.

Smaller compressor units are subject to a mechanical running test as well as to running-in of the pistons. A barring-over test is made on larger, completely assembled machines.

Test and material certificates are provided on request.

### "LABY'S" for liquefied natural gas service (low gas temperature application)

See Fig. C-116A, which shows the complete diagramatic assembly of compressor and accessories in liquefied natural gas (LNG) service. Figure C-117A is a photograph of the plant. Figure C-116B depicts the operating temperatures involved in a specific application. An LNG boil-off compressor has to cope with a variety of basic physical problems for which a product designed to normal standards would be inadequate. Two application aspects are of special interest in this context.

**Exposure to cryogenic temperatures.** LNG at barometric pressure boils off at –160°C. This temperature is well below the limit where some of the common engineering



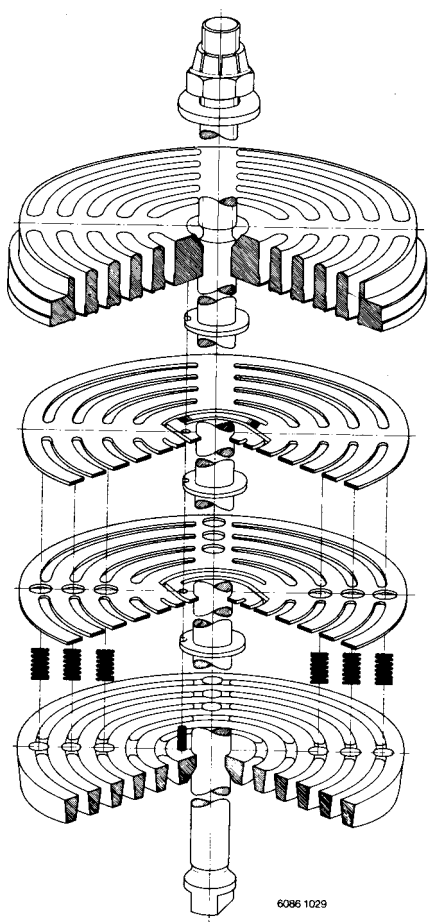



**FIG. C-107 The compressor valve. (Source: Sulzer-Burckhardt.)**

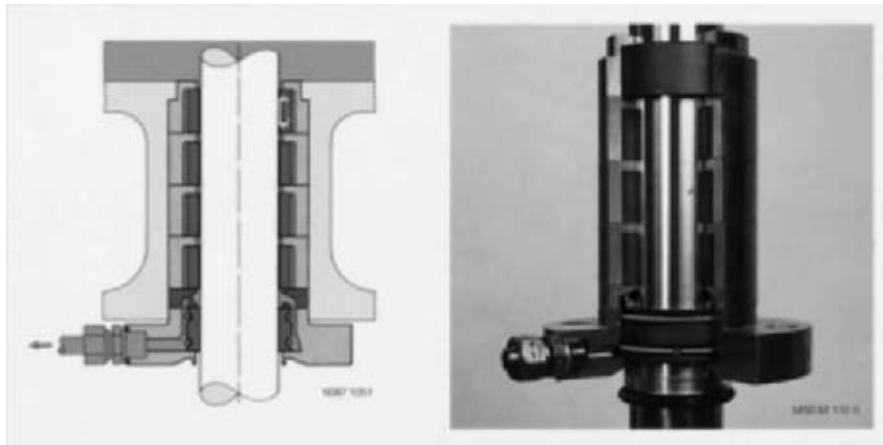


FIG. C-108 The piston rod gland. (Source: Sulzer-Burckhardt.)

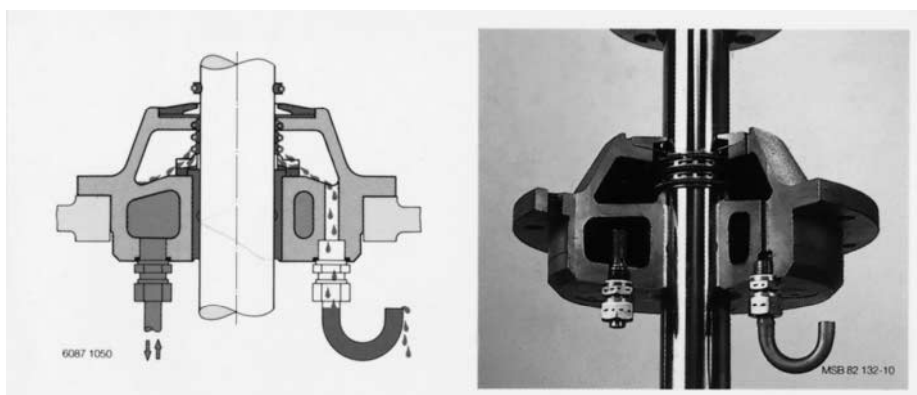



FIG. C-109 The piston rod guide bearing. (Source: Sulzer-Burckhardt.)

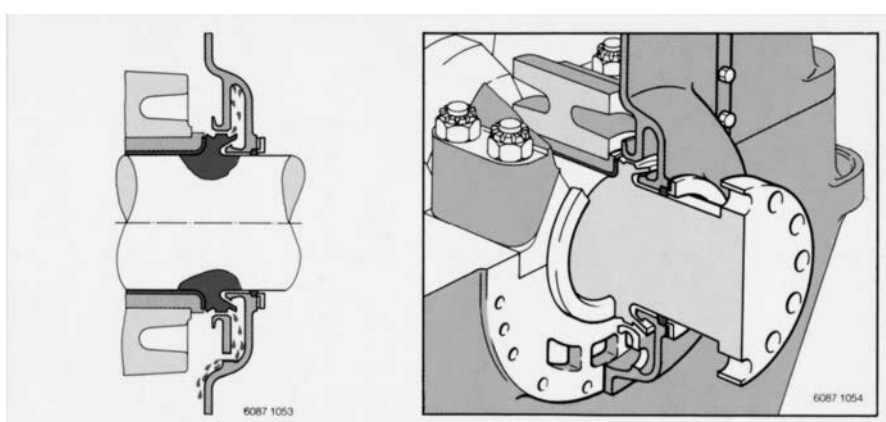



FIG. C-110 The crankshaft seal for open-type compressors. (Source: Sulzer-Burckhardt.)

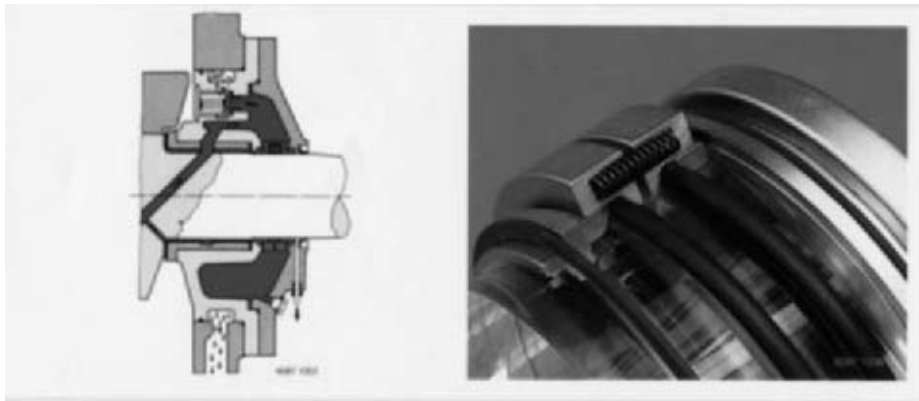
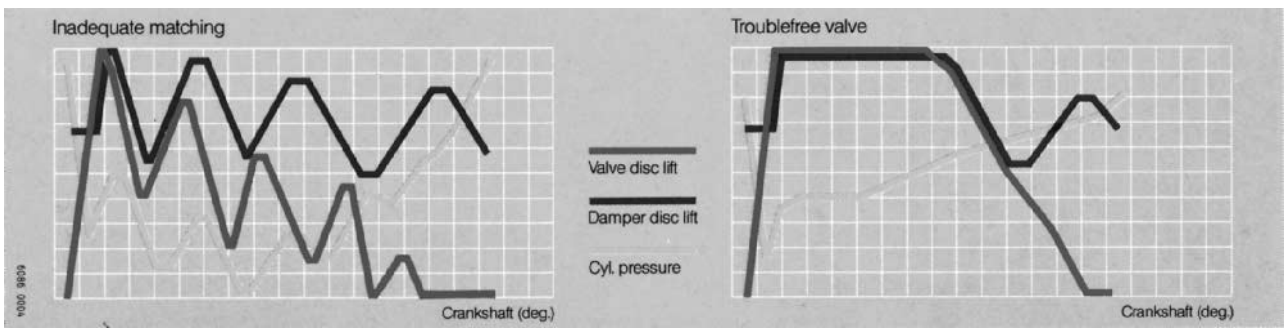


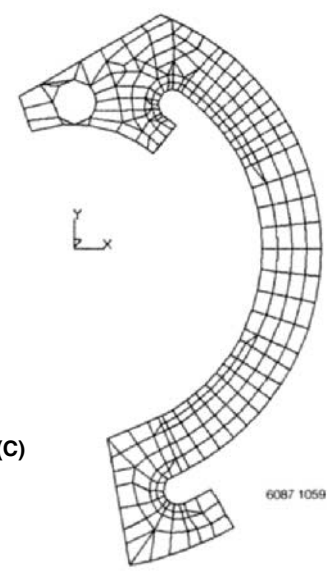
**FIG. C-111 The crankshaft seal for compressors with gas-tight crankcase. (Source: Sulzer-Burckhardt.)**




(A)



(B)




(C)




(D)

**FIG. C-112 Valve design troubleshooting: stress graphs and computer modeling. (Source: Sulzer-Burckhardt.)**

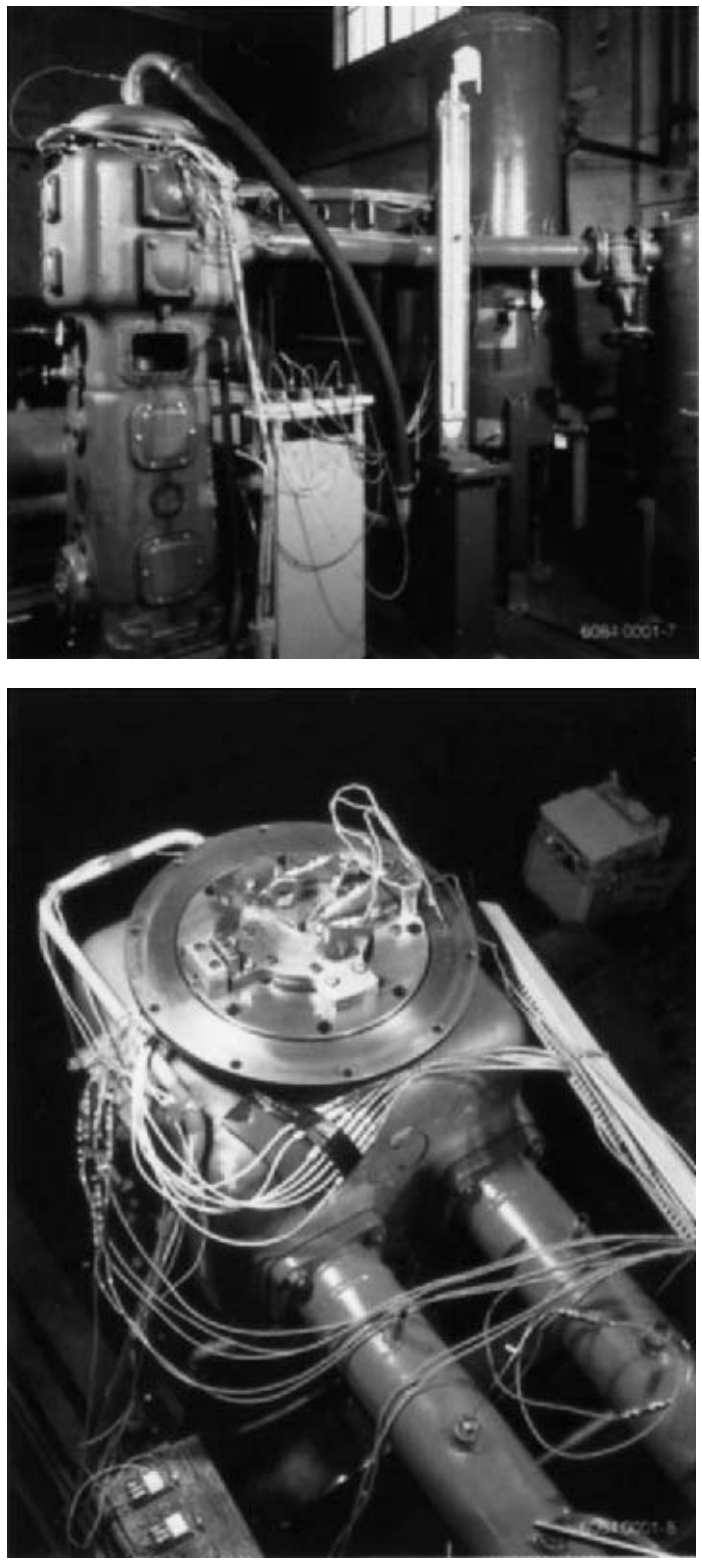


**FIG. C-113 Sealing labyrinth R&D rig. (Source: Sulzer-Burckhardt.)**

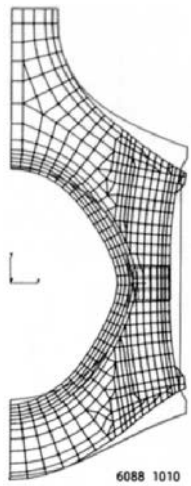



**FIG. C-114 Stress map taken during R&D on the dynamic crank. (Source: Sulzer-Burckhardt.)**

materials alter their properties. As an example, note the loss of ductility of most unalloyed carbon steels within a temperature span from 0°C to about –50°C. Figure C-117B shows the modes of impact transitions of C-steels and of Ni-alloyed nodular iron.

**Bone dry gas.** Natural gas in the form of boil-off is virtually free from water vapor as the dew point is as low as –160°C. Experience dictates to what extent moisture in a tribological system is an important parameter. Together with a number of other factors, it has a distinct bearing on wear rates under nonlubricated conditions. Those who decide to employ dry-running self-lubricating materials for piston rings must accept their mechanical and thermal constraints under bone dry running conditions. The operator must consequently set the stroke and speed of his machine in accordance with the gas conditions, so that the wear rate of the sealing and guiding elements can be held within acceptable limits. Already the initial choice of the dry-running material is itself subject to error because the designer is faced with a multitude of available material selections.

The operator is more free to optimize the design of individual parts of the compressor when the labyrinth principle is employed with the following main features:

- Avoidance of permanent mechanical friction
- Ability to use materials with known, easily certifiable qualities
- Simple design of the elements exposed to the process gas

**Design and material selection of pistons and cylinders (application case study).** One end-user installed a labyrinth piston compressor for handling LNG boil-off gas in a terminal and ran to 75,000 h quite inexpensively. Some end-users experience close or better time between overhauls (TBOs). See Table C-11.

The process data for the compressor in this application are as follows:

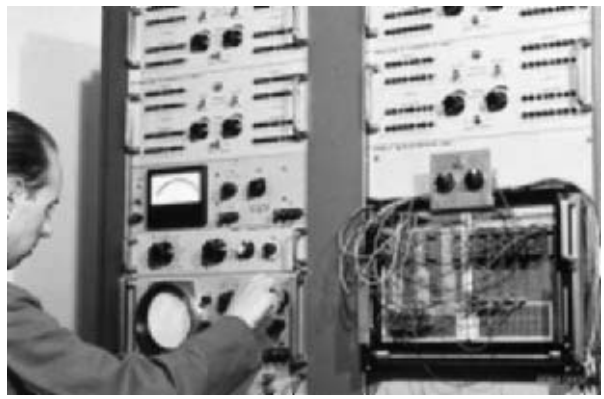




**FIG. C-115 Experimental rigs for R&D work. (Source: Sulzer-Burckhardt.)**

| Gas | | $CH_4$ (98%) + $N_2$ (2%) | | | | |
|---|---|---|---|---|---|---|
| Suction | 1st | 1.036 | bar | –90 | to | –160°C |
| Discharge | 1st | 5.2 | bar | +25 | to | –53°C |
| Suction | 2nd | 5.2 | bar | +25 | to | –53°C |
| Discharge | 2nd | 13.6 | bar | +38 | to | +102°C |
| Suction | 3rd | 13.6 | bar | +38 | to | +48°C |
| Discharge | 3rd | 23.4 | bar | +88 | to | +160°C |

**Material selection for cylinders and pistons.** The above data were the guideline for the materials selected for cylinders, labyrinth pistons, and other components of the machine. Figure C-118 illustrates a double-acting laby piston. The absence of tribological restrictions gained by using labyrinth sealing techniques allowed for the choice of the best-suited metals for the key components in each individual stage. For the 1st stage cylinders exposured to the lowest temperatures, the choice of GGG Ni35 was made. This is a nodular cast iron containing 35 percent nickel, also known under the trade name of Ni Resist D5. This alloy simultaneously exhibits remarkable ductility at low temperatures and one of the lowest thermal expansion coefficients known in metallurgy (Fig. C-119). The corresponding pistons were made of nickel-alloyed cast iron with laminar graphite.

Table C-12 indicates the outstanding thermal shock behavior of GGG Ni 35 in relation to other candidate materials. This is valuable, especially under transient

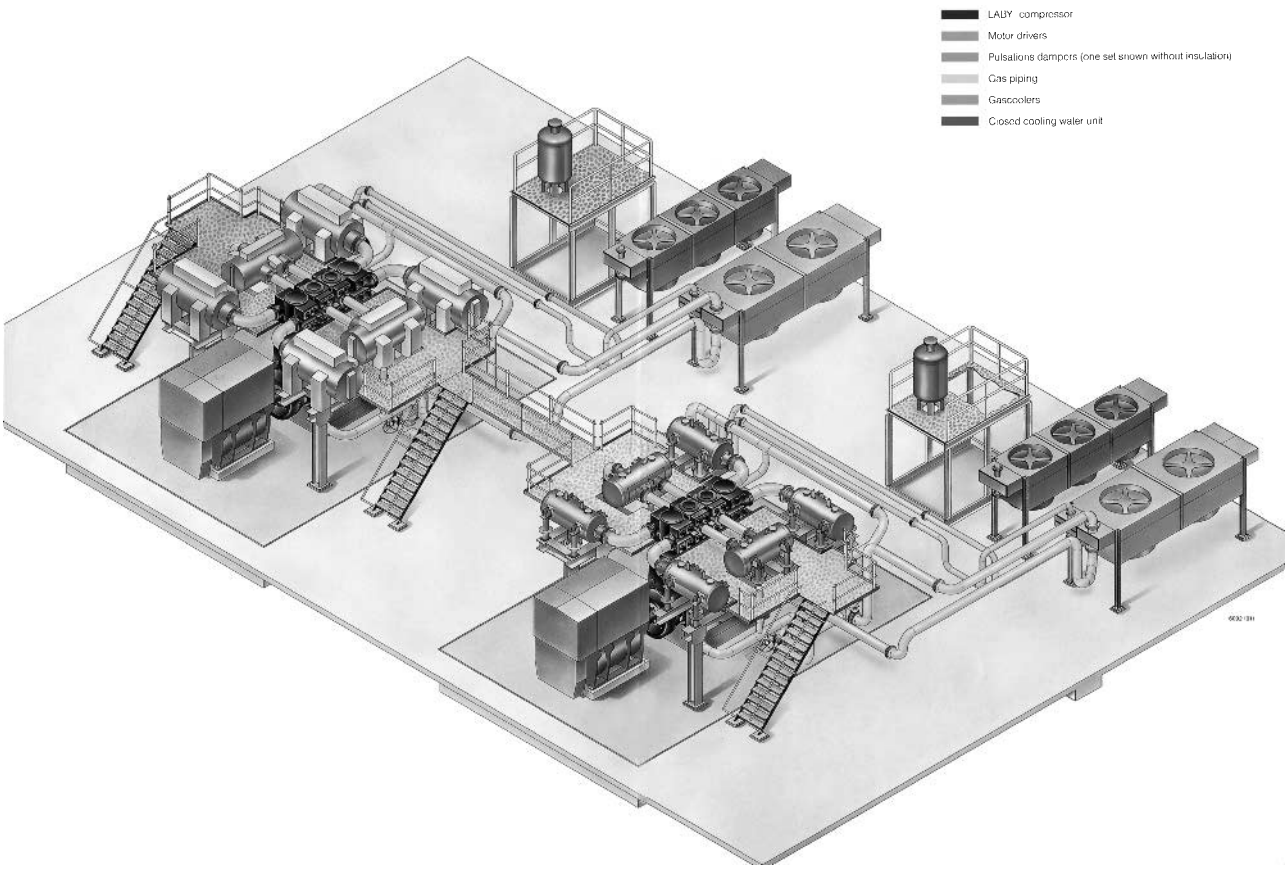



**FIG. C-116A Skid of compressor and accessories in LNG service. (Source: Sulzer-Burckhardt.)**

**TABLE C-11 Maintenance Report on a Period of 75,000 Running Hours**

| | | |
|---|---|---|
| Pistons (total of 4) | No replacement whatsoever | |
| Piston rod seals | 1st replacement after | 14,350 h |
| | 2nd replacement after | 36,993 h |
| | 3rd replacement after | 61,790 h |
| Piston rods | No replacement whatsoever | |
| Crankshaft seal | 1st replacement after | 14,350 h |
| | 2nd replacement after | 36,993 h |
| Bearings | | |
| Piston rods guide bearings | 1st replacement after | 36,993 h |
| | 2nd replacement after | 61,790 h |
| Crossheads | No replacement | |
| Crosshead pin bearings | No replacement | |
| Connecting rod bearings | No replacement | |
| Crankshaft bearings | One bearing lost after | 14,350 h |
| | No further replacement | |

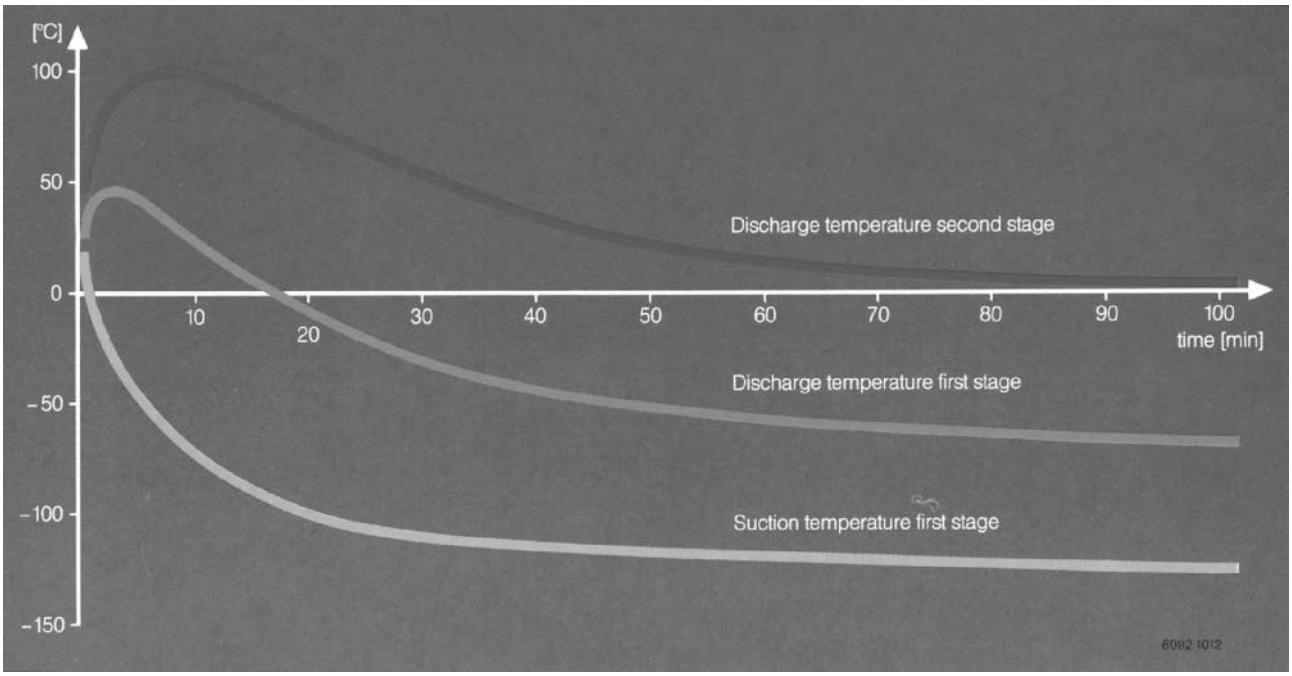



**FIG. C-116B The labyrinth-piston compressor is especially well adaptable to the great temperature drop from ambient to –150°C during startup. It is for this reason that no slow cooling-down procedure prior to startup is required. (Source: Sulzer-Burckhardt.)**



**FIG. C-117A Four labyrinth-piston compressors, type 3K 140-3C, in a storage facility in Sweden handling cold ethylene at –95 °C suction temperature from 1.01 bar abs to 31.5 bar abs in three stages. (Source: Sulzer-Burckhardt.)**

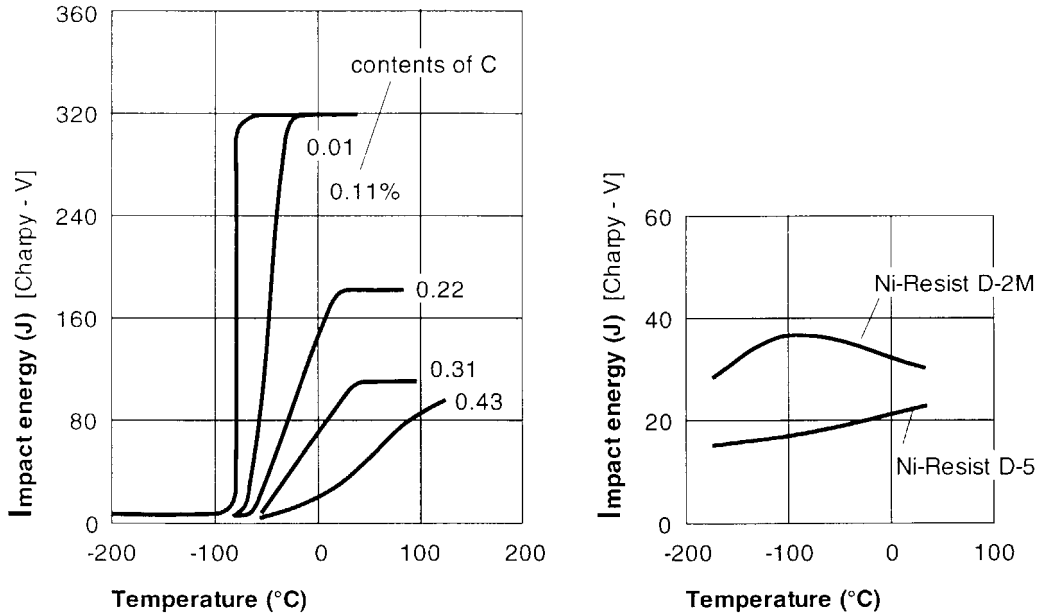



**FIG. C-117B Transitions of impact energies of nonalloyed C-steels and Ni-alloyed nodular iron ("Ni Resist"). (Source: Sulzer-Burckhardt.)**

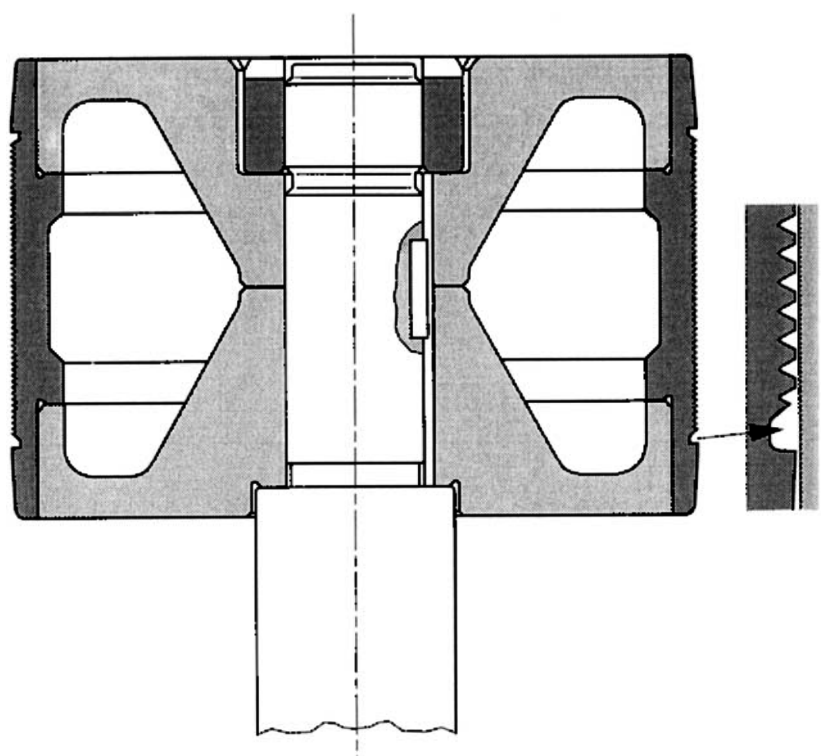


**FIG. C-118 Cutaway view of a double-acting labyrinth piston. (Source: Sulzer-Burckhardt.)**

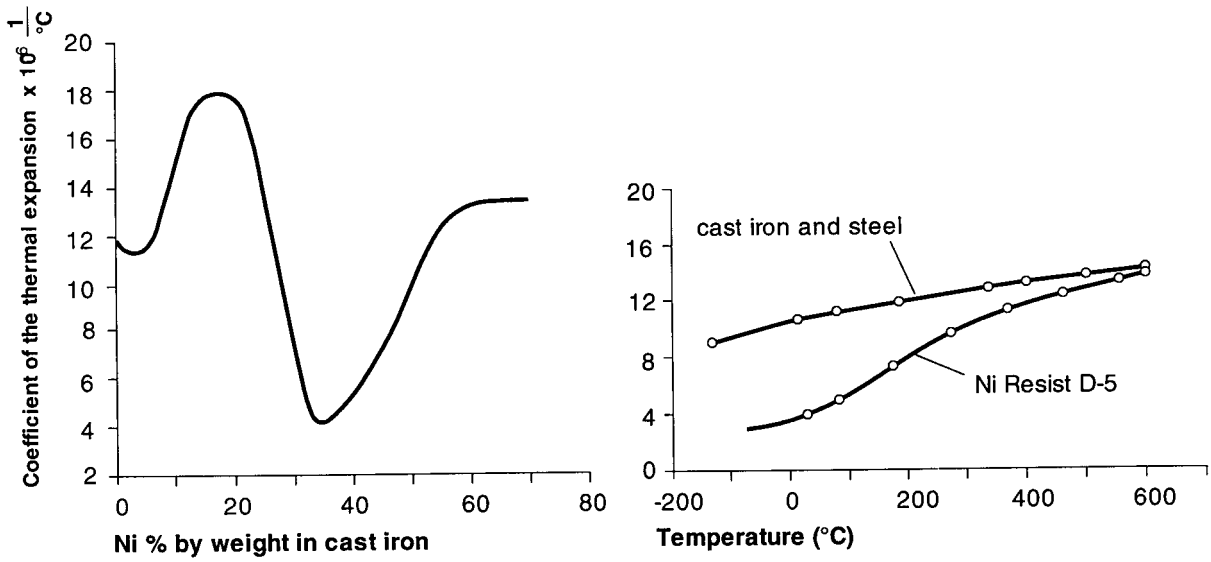



**FIG. C-119 Coefficient of thermal expansion for cast iron and steel in function of % Ni and temperature. (Source: Sulzer-Burckhardt.)**

**TABLE C-12 Candidate Materials for Low Temperature Components—Comparative Combined Properties**

| | ③ Tensile strength N/mm² | ① Endurance limit N/mm² | Youngs modulus N/mm² | Thermal expansion coefficient $10^{-6}$/°C | ② Thermal shock stress N/mm² (t = 100 °C) | ②/① Ratio | ②/③ Ratio |
|---|---|---|---|---|---|---|---|
| Cast iron GG 18 | 180 | 80 | 85,000 | 11.70 | 100 | 1.25 | 0.55 |
| Austenitic Steel CrNi | 460 : 600 | 230 : 300 | 204,000 | 20 | 410 | 1.80 : 1.40 | 0.89 |
| GGG NiCr 20 2 Type D2 | 430 | 190 | 125,000 | 17.6 | 220 | 1.10 | 0.51 |
| GGG Ni 35 Type D5 | 410 | 185 | 127,000 | 4.50 | 58 | 0.32 | 0.14 |

temperature conditions. It allows the compressor to be started directly of ambient temperature without any precooling. The less severe temperatures in the 2nd stage allowed the use of ferritic nodular cast iron with good fracture toughness down to –100°C and bronze for the piston. The 3rd stage cylinder consists of normal cast iron grade GG20.

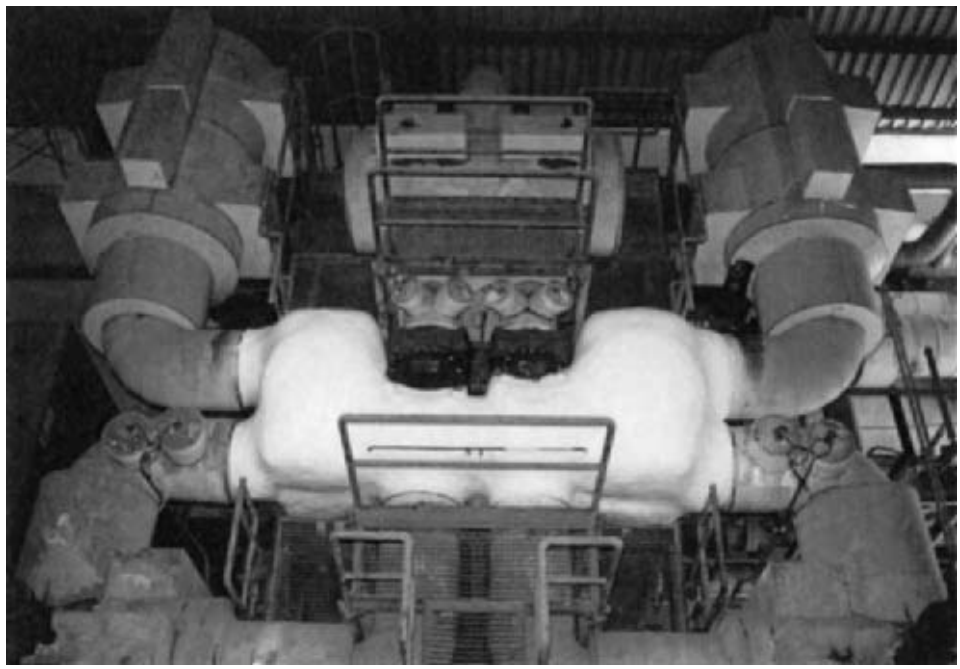


FIG. C-120 Icing of 1st stage cylinders after continuous operation of approximately 2–3 months. (Source: Sulzer-Burckhardt.)

**Control of temperature and deformation of the crankcase.** Gas temperatures at the 1st stage inlet valves are so low that energy imparted to the cylinders during gas compression raises their mean temperature to a value still well below that of ambient air. Therefore they do not have cooling jackets. They cool down well below the freezing point of the moisture in the natural atmosphere and consequently become covered with a thick layer of ice when the machine is running (Fig. C-120). To ensure a good alignment of the path of the labyrinth pistons, cold deformation of the crankcase underneath the 1st stage cylinders had to be prevented. This was achieved by means of a special water jacket that extends along the upper face of the crankcase and acts as a thermal barrier. See Fig. C-121 for component details.

**Internal and external leakage.** Consistent with the design of the pistons, labyrinth seals were also used between the double-acting cylinders and the distance piece at the upper end of the crankcase (Fig. C-122). Figure C-122 also shows details of the labyrinth seals around the piston rods. Each gland has a collector chamber before the lower end of the labyrinths from where the leak gas is internally returned to the suction upstream of the 1st stage cylinders.

To attain a perfect external tightness of the machine the passage of the crankshaft through the wall of the crankcase was sealed off by a rotating double-sided ring seal immersed in oil. Thus, the entire inside of the frame could be integrated into the gas-containing system and could be pressurized with either natural gas or an inert gas. In the case presented here it was left at suction pressure level and filled with natural gas. Figure C-123 shows details of the shaft seal. The entire machine represents, therefore, one hermetically closed shell with no gas leakage to the environment.

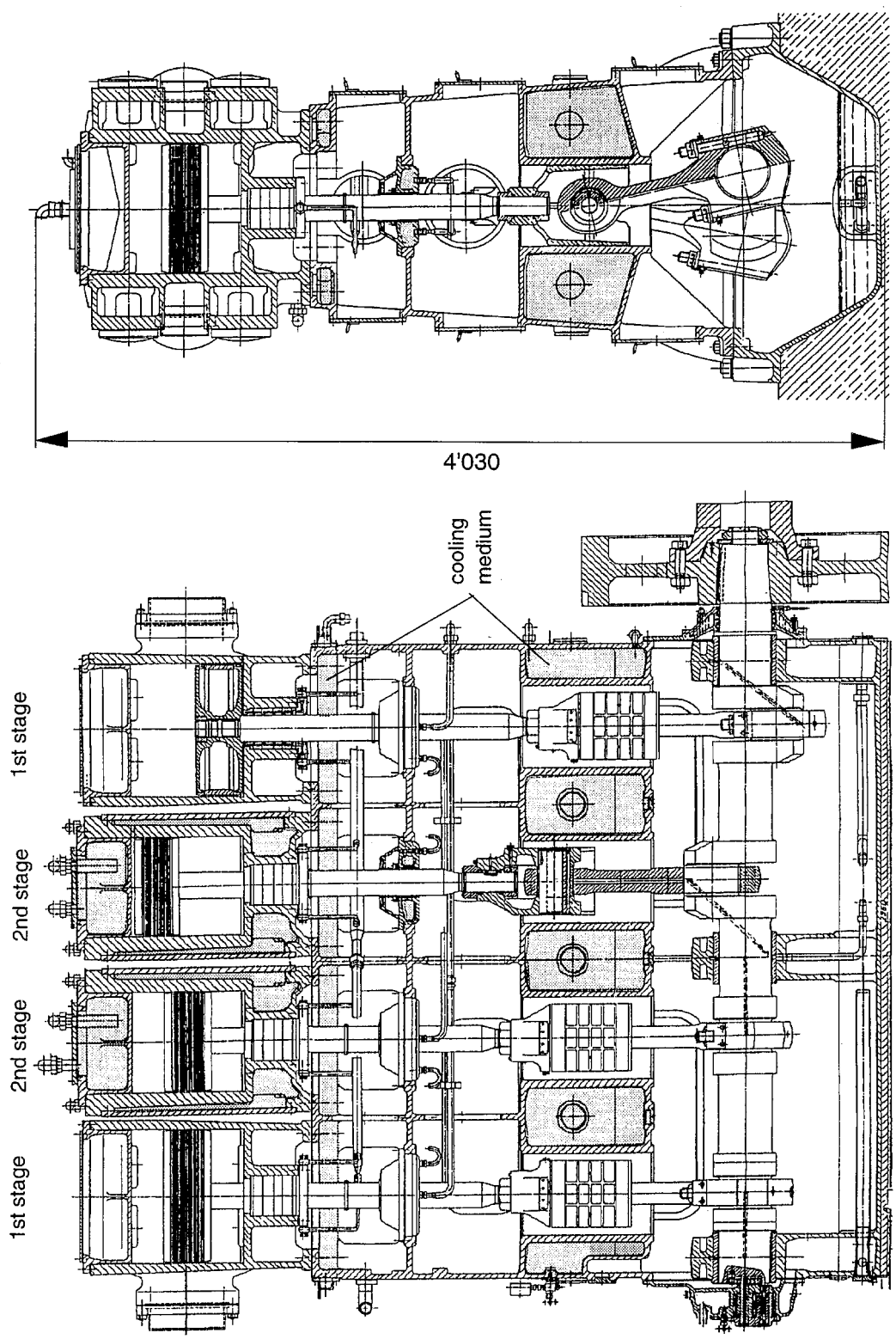



**FIG. C-121 Labyrinth-piston compressor for LNG boil-off: 4 double-acting cylinders; 3 compression stages; closed crankcasing; suction temperature −160°C. (Source: Sulzer-Burckhardt.)**

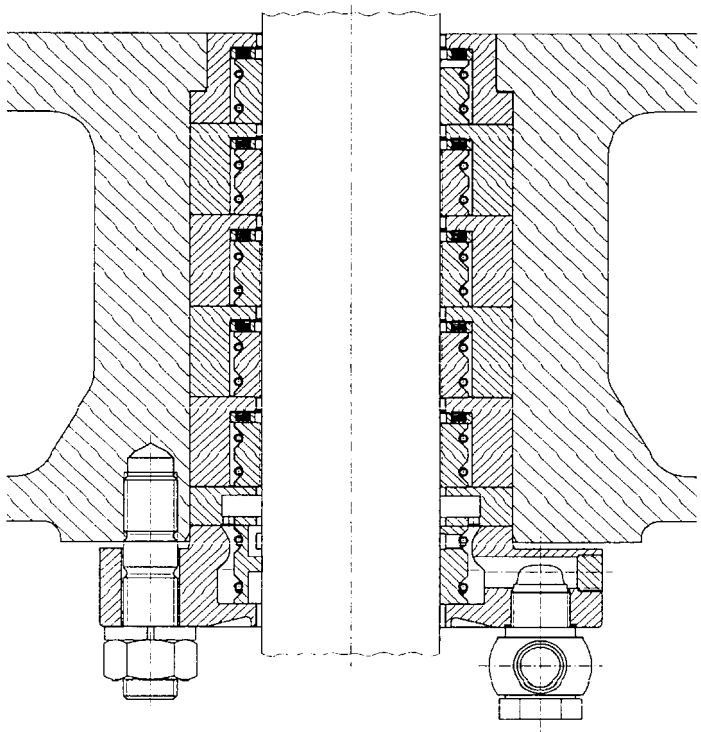


**FIG. C-122 Internal labyrinth sealing between double-acting cylinder and distance piece of crankcasing (piston rod sealing). (Source: Sulzer-Burckhardt.)**

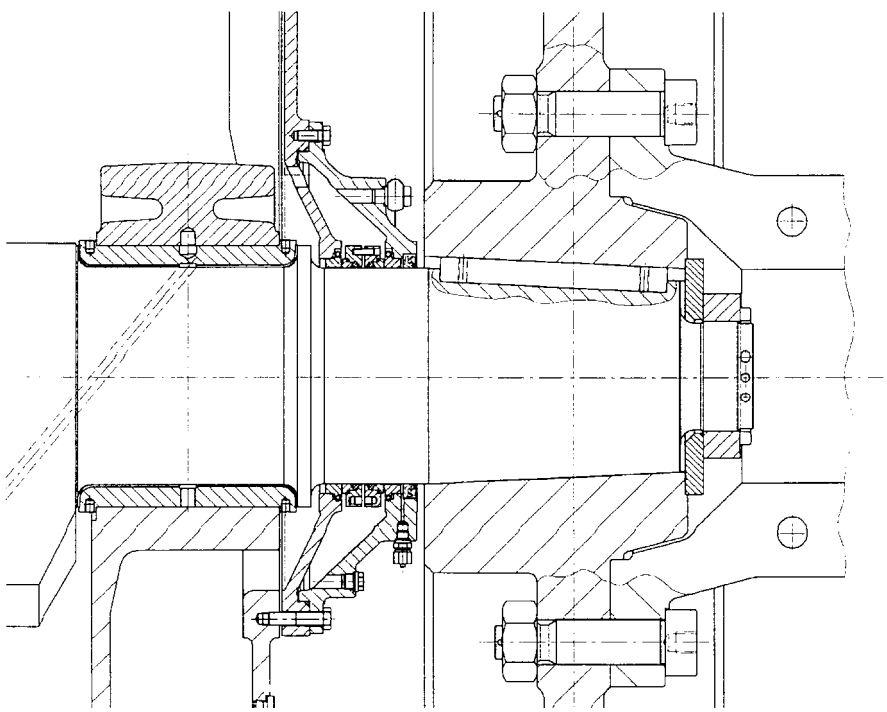


**FIG. C-123 Gas-tight sealing of crankshaft between crankcasing and environments. (Source: Sulzer-Burckhardt.)**

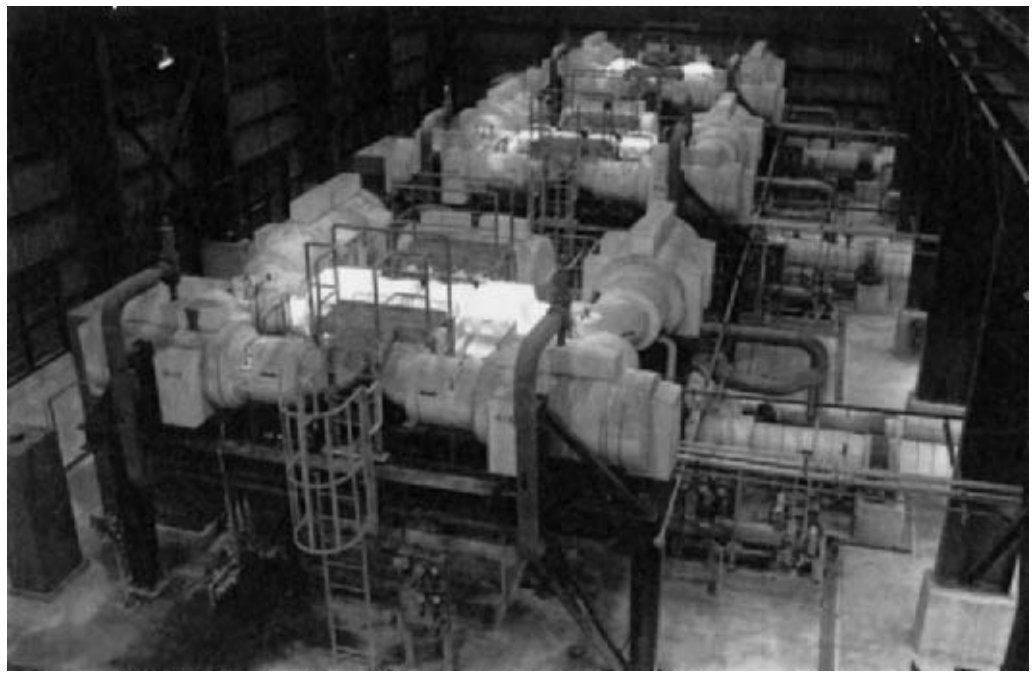


**FIG. C-124 Group of three labyrinth-piston compressors for LNG boil-off in a Taiwanese LNG receiving terminal. (Source: Sulzer-Burckhardt.)**

**Valves.** With valve discs, orders for replacment parts indicate an average life of at least 10,000 h with no changes based on running temperatures.

**Other LNG terminals.** The applications experience of the labyrinth piston compressor in this market segment has encouraged other terminals to install machines of the same kind. Figure C-124 shows a group of machines in a Taiwanese terminal where LNG is received, stored, and evaporated for distribution by a pipeline system throughout the island of Taiwan. When the terminal was extended in 1992, in view of the successful operation of these units an order for an additional compressor was placed. In this case a two-stage machine was required. As the discharge temperature after the 2nd stage reaches about −50 to 0°C only, both stages have uncooled cylinders.

The process data for this compressor are:

| Gas | | $CH_4$ (98%) + $N_2$ (2%) | | | | |
|---|---|---|---|---|---|---|
| Suction | 1st | 1.036 | bar | −106 | to | −160°C |
| Discharge | 1st | 2.9 | bar | −36 | to | −97°C |
| Suction | 2nd | 2.9 | bar | −33 | to | −93°C |
| Discharge | 2nd | 6.8 | bar | −45 | to | +35°C |

**Maintenance report on these compressor applications.** By the end of January 1996 the first three compressors had all passed 15,000 running hours. One compressor was opened up after 10,000 running hours. Piston and bearing clearances, piston rod seals, and valves were checked. Some gland rings and some minor valve parts were replaced. The maintenance report read: "like new!"

**Startup procedure.** Special attention was given to developing a simple startup procedure. The low thermal expansion coefficient of the chosen cylinder material

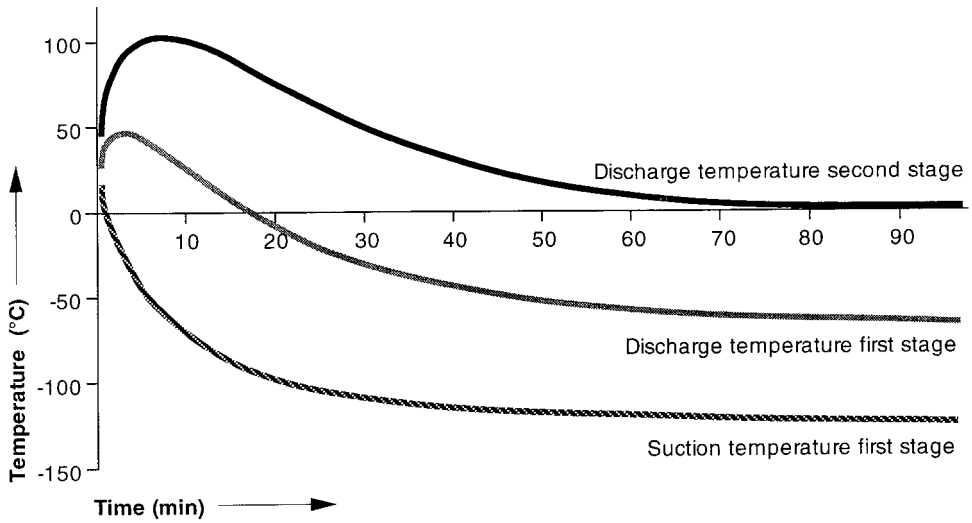



**FIG. C-125 Gas temperature readings taken during the startup of a LNG boil-off gas labyrinth-piston compressor. (Source: Sulzer-Burckhardt.)**

leads to low thermal stress. Together with carefully designed pulsation dampeners and a well-engineered gas piping arrangement, the system allows full automatic startup of the compressors. The transition from ambient temperature down to boil-off temperature is achieved without any precooling (Fig. C-125). Figure C-126 shows two other LNG boil-off gas compressors in the design stage commissioned in a Korean LNG receiving terminal.

**Summary of operational experience thus far.** Low gas temperature challenges gas compressors in two ways:

- Physical contact with cold gas and consequences for material properties
- Absence of humidity (low dew point) with a strong bearing on tribology in nonlubricated areas

The application of labyrinth seals in reciprocating compressors is a logical answer to these problems. Labyrinth piston compressors have demonstrated this in industrial operation successfully down to boil-off temperature of natural gas at –160°C. Such machines can be built with zero leakage to the environment. They need little maintenance. Plant operators and maintenance crews claim low attention requirements.

### Reference and Additional Reading

1. Bloch, H., and Soares, C. M., *Process Plant Machinery*, 2d ed., Butterworth-Heinemann, 1998.

## Turbocompressors (Includes Centrifugal, Axial, and Isothermal Compressors, as Well as Fans)

### General Information*

A diagrammatic summary of operating ranges of various categories under this subdivision is presented here, followed by further details and information on some of the more important compressor categories. The model designations are specific to this OEM information source. When selecting a compressor, one way of not

*Source: Sulzer-Burckhardt, Switzerland. Adapted with permission.

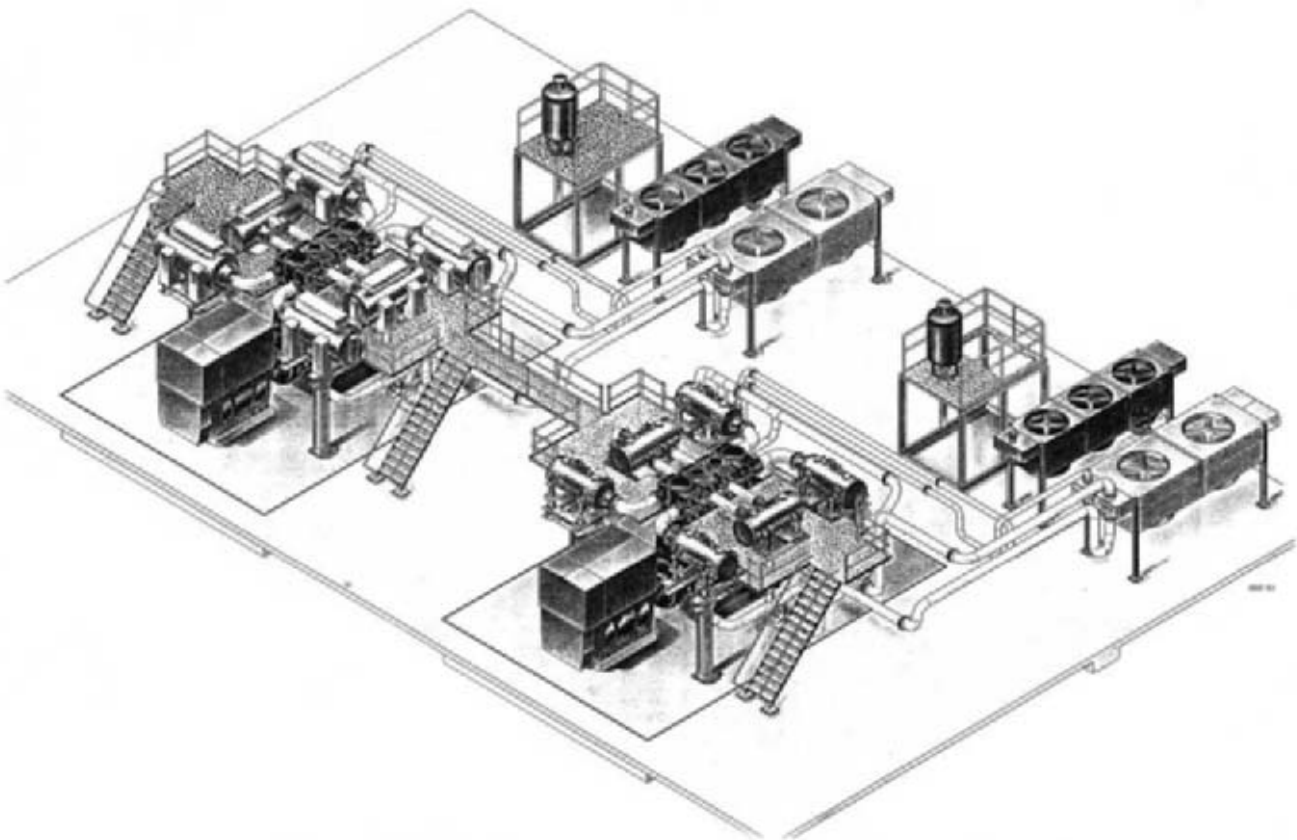


FIG. C-126 Two identical labyrinth-piston compressor plants, put into operation in a Korean LNG receiving terminal in December 1993. (Source: Sulzer-Burckhardt.)

omitting any vital features is to consider the specifications of one well-known OEM and compare other OEMs for equivalent features.

The pressure-volume diagram (Fig. C-127) shows the duty range of the standard model series for this OEM source. Each compressor's size is arrived at with different numbers of stages and nominal pressures, depending on the application. Pictorial operational summaries of these main categories are shown in Figs. C-129 through C-141.

**Turbocompressor controls (*see* Control Systems)**

**Axial compressors***

Axial compressors are being used increasingly for applications that not long ago were considered the domain of centrifugal machines. Thanks to their high specific flow capacity, the corresponding low weight, reduced space requirement, and particularly their high efficiency, the axial compressors play a major role in the reliable and economic operation of modern, large-scale industrial plants. They now form a vital and indispensable integral part of installations like blast furnaces, air separation plants, fluid catalytic cracking units, nitric acid plants, jet-engine test facilities, thermocompression, LNG, and synfuel processes. See Fig. C-128.

*Source: Sulzer-Burckhardt, Switzerland. Adapted with permission.

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